

Q1: Average GRE of accepted PhD applicants

Answer: The average GRE score of accepted PhD applicants is 177.95 (based on available GRE Quant scores).

Q2: Average GPA of accepted Master's applicants

Answer: The average GPA of accepted Master's applicants is 4.09 (based on 3,830 applicants with valid GPAs).

Q3: Shortcomings of inferring decision_date from text status and term

Answer:

1. Inconsistent formatting: The decision_date field appears in multiple text formats (e.g., "on 15 Feb", "on May 23", "on Apr 23"), making parsing difficult and error-prone.
 2. Missing data: Many rows have no decision date at all, making it impossible to infer a full date.
 3. Term inference may be inaccurate: Inferring the year from the term field assumes that the decision date falls within the same academic year as the term, which may not always be true for late decisions, rolling admissions, or applicants who defer their enrollment.
-

Q4: Correlation analysis between GPA and GRE-related measures

Answer:

- GPA vs GRE correlation: -0.0595 ($p = 0.0106$) — This indicates a very weak negative correlation that is essentially negligible. GPA and GRE scores appear to be largely independent of each other.
 - GRE vs GRE V correlation: 0.2175 ($p = 0.0000$) — This indicates a weak but statistically significant positive correlation between GRE Quant and GRE Verbal scores.
-

Q5: Group comparison — Accepted vs Rejected GPA

Answer: The group comparison did not suggest a significant difference in GPA between accepted and rejected applicants.

- Accepted applicants had an average GPA of 3.79.
 - Rejected applicants had an average GPA of 3.81.
 - The t-test produced a p-value of 0.2908, which is greater than the conventional significance level of 0.05. Therefore, we cannot reject the null hypothesis that the two groups have the same average GPA.
-

Q6: Chi-square test — Degree vs Outcome

Answer: The chi-square test between Degree and Outcome yielded:

- Chi-square statistic: 233.53
- p-value: 0.0000

This indicates a statistically significant association between degree type and admission outcome. The contingency table shows:

- Master's applicants: 6,200 accepted, 4,866 rejected
- PhD applicants: 4,139 accepted, 5,007 rejected
- PsyD applicants: 1 accepted, 0 rejected

Master's applicants appear to have a higher acceptance rate compared to PhD applicants.

Q7: Remaining data quality issues

Answer: Several data quality issues remain in the cleaned dataset:

- Missing GPA: 55.7% of rows have missing GPA values.
- Missing GRE Quant: 73.7% of rows have missing GRE Quant scores.
- Missing GRE Verbal: 74.9% of rows have missing GRE Verbal scores.
- Missing GRE AW: 75.9% of rows have missing GRE AW scores.
- Missing outcome: 25.8% of rows have missing outcome values (Waitlisted/Interviewed/Rejected/Accepted).

These high missing rates limit the statistical power of analyses involving these variables and suggest that future data collection efforts should focus on improving the completeness of academic metrics.